

Feature Design Document: Customizable Inventory & Service Management Platform

1. Introduction

This document outlines the detailed feature design for a customizable inventory and service management platform, aiming to surpass existing solutions like Vyapar App. The design incorporates core functionalities, addresses specific pain points identified by the user, and leverages the foundational work already present in the MobiManager GitHub repository. The primary goal is to create a flexible platform where users can select and pay only for the features they need, catering to diverse electronics businesses such as mobile shops, laptop repair centers, and appliance sellers.

2. Vyapar App Feature Analysis

Vyapar App is a popular GST billing, accounting, and inventory management software for small businesses in India. Its key features include:

- **GST Billing & Invoicing:** Generate GST-compliant invoices quickly (under 20 seconds) and share them via WhatsApp, email, or print in various formats (A4, A5, 2", 3" thermal). It offers over 10 professional invoice themes and auto-calculates CGST, SGST, and IGST. The app also stores invoices for future reference, audit, and GSTR filing. It supports both online and offline operations, which is crucial for businesses in areas with unreliable internet connectivity 1.
- **Inventory Management:** Provides a live view of stock levels across stores or warehouses, with automatic updates on sales and purchases. Features include low-stock alerts, batch tracking, expiry date management, and slot number tracking. It helps in identifying slow-moving inventory through sales trend reports 1.
- **Payment Reminders:** Automated due-payment alerts can be sent to customers via WhatsApp or SMS, helping businesses recover dues efficiently 1.
- **GST Filing:** The software auto-generates various GST returns, including GSTR-1, GSTR-2, GSTR-3B, and GSTR-9, simplifying tax compliance 1.
- **Business Reports:** Offers more than 50 types of business reports, such as Balance Sheets, Profit & Loss statements, sales reports, and purchase reports, providing comprehensive financial insights 1.
- **Multi-Device Sync:** Allows seamless synchronization of data between mobile and desktop applications, enabling business owners to monitor activities from anywhere 1.

- **User Management:** Includes a basic user management feature, likely with role-based access, though specific details on granularity are not extensively highlighted on their public pages 1 .
- **Online Business Store:** Enables businesses to create a digital catalog, allowing customers to place orders online 1 .
- **Expense Tracking:** Facilitates recording and managing business expenses to maintain accurate financial records 1 .
- **Estimates & Quotations:** Users can easily create and send estimates and quotations, which can then be converted into invoices 1 .
- **Order Tracking:** Manages both sales and purchase orders, providing a clear overview of order statuses 1 .
- **Delivery Challan:** Supports tracking of delivery challans for goods in transit 1 .
- **Bank Account Linking:** Allows linking bank accounts for reconciliation and tracking transactions 1 .
- **Barcode Scanning:** Supports adding barcodes to items and scanning them during sales and purchase processes for efficient inventory management 1 .

Vyapar offers a free mobile app for Android with core billing features, while premium features and desktop access are available through an affordable annual subscription 1 .

3. MobiManager Repository Analysis

The user's MobiManager repository provides a strong foundation for the proposed platform, with several key architectural and functional components already in place or planned. The `prisma/schema.prisma` file reveals the following data models and intended functionalities 2 :

- **Authentication & Authorization:** Implemented with Super Admin and Admin login using JWT tokens, bcrypt for password hashing, and middleware-based route protection. A robust Admin verification workflow is in place, requiring document uploads (Aadhaar, PAN, Shop Act) and SuperAdmin review 2 .
- **Multi-Tenancy:** The schema supports multiple `Shop` entities under a single `Admin` , addressing the user's requirement for multi-shop management. `SubAdmin` roles are also defined, indicating support for granular user permissions 2 .
- **Subscription & Plans:** A `Plan` model exists, defining subscription tiers (Starter, Pro, Elite) with attributes like monthly/yearly pricing, maximum products, sub-admins, and shops. This provides a basis for a customizable plan structure 2 .

- **Inventory Management:** `Product` and `StockMovement` models are present, allowing for tracking of inventory items, purchases, sales, returns, and adjustments. Categories like `MOBILE` and `ACCESSORY` are already defined 2 .
- **Sales & Transactions:** `Sale` and `SaleItem` models manage customer transactions, while `RechargeTransfer` handles service-based transactions (Mobile Recharge, DTH, Electricity, Money Transfer) 2 .
- **Repair Management:** A `Repair` model is included with statuses such as `RECEIVED` , `IN_REPAIR` , `REPAIRED` , `DELIVERED` , and `CANCELLED` . This forms a good starting point for detailed repair tracking 2 .
- **Audit Logging:** An `AuditLog` model is defined to track actions performed by Admins and SubAdmins, enhancing security and accountability 2 .
- **Technical Stack:** The project uses Next.js 14, TypeScript, PostgreSQL with Row Level Security, Prisma ORM, Docker, and Tailwind CSS, indicating a modern and scalable architecture 2 .

4. Proposed Feature Design

Building upon the strengths of MobiManager and addressing the gaps identified in Vyapar, the following features are proposed for the customizable inventory and service management platform:

4.1. Core Business Management

- **Customizable Dashboard (अनुकूलन योग्य डैशबोर्ड):** A personalized dashboard for each Admin and SubAdmin, displaying relevant metrics and quick actions based on their roles and subscribed features. This will provide an at-a-glance overview of sales, inventory, repairs, and other critical business data.
- **GST Billing & Invoicing (जीएसटी बिलिंग और चालान):** Enhanced GST-compliant invoicing with advanced customization options for templates, logos, and terms. Support for multiple tax rates, discounts, and payment terms. Instant sharing via WhatsApp, email, and print. (Similar to Vyapar, but with more flexibility).
- **Expense Management (खर्च प्रबंधन):** Comprehensive tracking of all business expenses, categorized for better financial analysis. Ability to attach receipts and generate expense reports.
- **Banking & Reconciliation (बैंकिंग और समाधान):** Seamless integration with multiple bank accounts for automatic transaction import and reconciliation. This will provide a real-time view of cash flow.

- **Reporting & Analytics (रिपोर्टिंग और विश्लेषण):** Over 100 customizable business reports, including sales trends, profit & loss, balance sheets, stock movement, customer purchase history, and repair analytics. Export options to various formats (PDF, Excel).

4.2. Advanced Inventory Management

- **Granular Product Variants (उत्पाद वेरिएंट का विस्तृत प्रबंधन):** This is a critical feature for mobile shops. Instead of a single product entry, allow defining product variants based on attributes like:

- **Mobile Model:** (e.g., iPhone 15, Samsung Galaxy S24)
- **Accessory Type:** (e.g., Back Cover, Screen Protector, Charger)
- **Color:** (e.g., Black, Blue, Clear)
- **Material:** (e.g., Silicone, Leather, Hard Plastic)
- **Storage:** (e.g., 128GB, 256GB)
- **Condition:** (e.g., New, Used, Refurbished)

Each unique combination will have its own SKU, stock level, and pricing. This addresses the user's pain point of managing mobile covers (e.g.,

iPhone 15 Pro Max, Silicone, Blue cover). This can be extended to other electronics like laptops (e.g., Dell XPS 15, i7, 16GB RAM, 512GB SSD).

- **Barcode & QR Code Generation/Scanning (बारकोड और क्यूआर कोड जनरेशन/स्कैनिंग):** Integrated tools for generating and printing barcodes/QR codes for each product variant. Fast and accurate scanning for sales, purchases, and stock adjustments using mobile devices or dedicated scanners.
- **Multi-Warehouse/Shop Inventory (मल्टी-वेयरहाउस/शॉप इन्वेंटरी):** Real-time stock visibility across all linked shops/warehouses. Ability to transfer stock between locations and track inventory in transit. This directly supports the user's multi-shop requirement.
- **Stock Alerts & Reorder Points (स्टॉक अलर्ट और रीऑर्डर पॉइंट):** Advanced customizable alerts for low stock, expiring products, and slow-moving items. Automated reorder suggestions based on sales velocity and supplier lead times.
- **Batch & Serial Number Tracking (बैच और सीरियल नंबर ट्रैकिंग):** Essential for electronics. Track individual items by their unique serial numbers for warranty management, recall tracking, and detailed sales history. Batch tracking for accessories.

4.3. Enhanced Service & Repair Management

- **Comprehensive Repair Workflow (व्यापक मरम्मत कार्यप्रवाह):** Expand the existing Repair model to include:

- **Detailed Job Sheets:** Capture customer details, device information (make, model, serial number, IMEI), reported issue, diagnostic notes, estimated cost, and estimated completion time.
- **Status Tracking:** Beyond RECEIVED , IN_REPAIR , REPAIRED , DELIVERED , CANCELLED , add statuses like AWAITING_PARTS , READY_FOR_DELIVERY (addressing the user's 'repaired but not delivered' pain point), CUSTOMER_NOTIFIED , PAYMENT_PENDING .
- **Parts Management:** Link repair jobs to StockMovement for parts used, automatically updating inventory. Ability to order specific parts for repairs.
- **Technician Assignment & Tracking:** Assign repair jobs to specific technicians and track their progress and performance.
- **Customer Communication:** Automated SMS/WhatsApp updates to customers on repair status changes.
- **Service History (सेवा इतिहास):** Maintain a complete service history for each device, including past repairs, parts used, and technician notes.

4.4. Customizable Plans & Pricing (अनुकूलन योग्य योजनाएँ और मूल्य निर्धारण)

This is a core differentiator. The platform will move beyond fixed plans to a modular, feature-based subscription model.

- **Base Plan (मूल योजना):** A free or very low-cost base plan with essential billing and basic inventory features.
- **Feature Modules (फीचर मॉड्यूल):** Each advanced feature will be a separate, purchasable module. Examples:
 - Advanced Inventory Management (Granular Variants, Batch/Serial Tracking, Multi-Warehouse)
 - Enhanced Repair Management (Detailed Workflow, Parts Management, Technician Tracking)
 - Advanced Reporting & Analytics
 - Multi-Shop Management
 - Multi-SubAdmin Management (beyond a basic limit)
 - Online Storefront
 - API Access for Integrations
- **Usage-Based Add-ons (उपयोग-आधारित ऐड-ऑन):** For certain features, pricing can be usage-based (e.g., number of SMS notifications, number of active SubAdmins beyond a certain limit, storage for documents).

- **Custom Plan Builder (कस्टम प्लान बिल्डर):** A user-friendly interface where Admins can select desired features and add-ons, and the system dynamically calculates their monthly/yearly subscription cost. This directly addresses the user's request for a custom plan design where they pay only for what they use.

4.5. User Management & Permissions (उपयोगकर्ता प्रबंधन और अनुमतियाँ)

This section will significantly enhance the existing Admin and SubAdmin models to provide granular control.

- **Multi-Shop Management (मल्टी-शॉप प्रबंधन):** An Admin (shop owner) can create and manage multiple Shop entities under a single account. Each shop will have its own inventory, sales, and repair data, but accessible and manageable from a central Admin dashboard.
- **Multi-SubAdmin Support (मल्टी-सबएडमिन सपोर्ट):** Admins can create multiple SubAdmin accounts for each shop or across multiple shops. This allows for delegation of tasks to employees.
- **Granular Role-Based Access Control (RBAC) (बारीक भूमिका-आधारित पहुँच नियंत्रण):** The most crucial enhancement. Admins will be able to define custom roles and assign specific permissions to each SubAdmin. This goes beyond simple predefined roles and allows for extreme flexibility. Examples of permissions:
 - **Module Access:** (e.g., Can access Inventory, Can access Repairs, Can access Sales).
 - **Action Permissions:** (e.g., Can Create Invoice, Can Edit Invoice, Can Delete Invoice, Can View Reports, Can Add Product, Can Edit Product, Can Delete Product, Can Change Repair Status).
 - **Data Scope:** (e.g., Can only view/manage inventory for Shop A, Can view sales for all shops).
 - **Financial Permissions:** (e.g., Can view profit/loss, Can access banking details, Can process refunds).
 - **Sensitive Actions:** (e.g., Can change pricing, Can give discounts, Can approve returns).
- **Activity Logs (गतिविधि लॉग):** Enhanced audit logs to track every action performed by Admins and SubAdmins, including timestamps, IP addresses, and specific changes made. This ensures accountability and security.
- **User ID & Password Management:** Admins will have full control over creating user IDs and passwords for their SubAdmins, including forcing password resets and setting password policies.

4.6. Other Specialized Features

- **Recharge & Money Transfer Management (रिचार्ज और मनी ट्रांसफर प्रबंधन):** Building on the existing `RechargeTransfer` model, provide a dedicated interface for managing various service-based transactions (mobile recharges, DTH, electricity bill payments, money transfers). Track commissions, service charges, and transaction history.
- **Customer Relationship Management (CRM) (ग्राहक संबंध प्रबंधन):** Basic CRM functionalities to manage customer profiles, purchase history, repair history, and communication logs. This helps in personalized service and marketing.
- **Supplier Management (आपूर्तिकर्ता प्रबंधन):** Track supplier details, purchase history, payment terms, and outstanding payables.
- **Integrated Payment Gateway (एकीकृत भुगतान गेटवे):** Integration with popular Indian payment gateways (e.g., Cashfree, Razorpay) for seamless collection of payments from customers and disbursement to suppliers.
- **Offline Mode with Sync (ऑफलाइन मोड के साथ सिंक):** Robust offline capabilities for billing and inventory updates, with automatic synchronization once an internet connection is restored. This is vital for businesses in areas with intermittent connectivity.

5. Technical Considerations

- **Scalability:** The chosen tech stack (Next.js, TypeScript, PostgreSQL, Prisma, Docker) already provides a scalable foundation. Emphasis will be on optimizing database queries and API performance to handle a growing user base and data volume.
- **Security:** Continue to prioritize security with Row Level Security, Zod validation, Helmet security headers, and rate limiting. Regular security audits and updates will be crucial.
- **User Interface/User Experience (UI/UX):** A clean, intuitive, and responsive UI/UX will be paramount. The design will focus on ease of use, especially for complex tasks like granular inventory management and repair workflows. The interface will be available in both Hindi and English.

6. Conclusion

This detailed feature design document outlines a comprehensive and highly customizable inventory and service management platform that aims to address the specific needs of electronics shop owners in India. By offering granular control over features, multi-shop and multi-subadmin management, and specialized inventory tracking for items like mobile covers, the platform will provide a superior alternative to existing solutions. The modular

pricing model will empower users to build a plan that perfectly fits their business requirements, ensuring they pay only for the functionalities they truly utilize.

References

[1] Vyapar App. Best GST Billing Software for Small Businesses | Vyapar.

[2] Aveenod. MobiManager.